Software Project Management (SPM) – Semester Project Proposal

Clinic Appointment Management System

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Abstract

This document demonstrates the expected Week 3 Overleaf output for an Clinic Appointment Management Application. It includes a project proposal,requirements, three UML diagrams,numbered figures, an equation, cross-references, and cited sources.

1 Project Overview

1.1 What: Project Description

A web-based system that allows patients to book, reschedule, and cancel appointments with doctors online, while clinic staff can manage doctor schedules, patient queues, and appointment records efficiently.

1.2 Why: Problem Statement

Currently, most clinics rely on phone calls or walk-in serials, causing long waiting times,double bookings and scheduling conflicts. This project solves those problems by providing a centralized, transparent appointment booking process that reduces crowing and improves patient experience.

1.3 How: Project Plan

We will follow structured SPM approach:

1. Prepare a Software Requirements Specification(SRS) based on stakeholder needs
2. Create a Work Breakdown Structure (WBS) and Gantt chart for scheduling
3. Estimate costs (effort, tools, and resources) using a suitable model
4. Develop a risk register with mitigation plans (e.g., data privacy, schedule delays)
5. Define a quality/test plan and documentation plan
6. Produce a final project management plan and presentation

1.4 Management Approach

Progress is tracked with a shared task board. The team uses Git for version control and Overleaf for documentation. A short stand-up meeting is held twice a week.

1.5 Who: Targets Users

- Primary users: Patients (book appointments)
- Secondary users: Doctors (view schedules), Receptionists/Clinic staff (manage bookings,queues)
- Stakeholders: Clinic owner/administrator, IT support, and project team

1.6 Expected Outcomes:

- A complete SPM project plan including SRS, WBS, schedule, cost estimate, risk register, and test plan
- A clear understanding of how to manage a real-world software project from initiation to closure
- A final presentation and one-page proposal as required

1.7 Team Members and Roles

Name	Role	Responsibility
Numaier Islam Khan	Project Manager and Documentation Lead	Overall team coordination, stakeholder communication, schedule monitoring, meeting minutes, final proposal/report, and presentation preparation
Shahedul Alam	Business Analyst and Risk/Quality Lead	Requirements gathering, SRS and user stories preparation, stakeholder analysis, risk register and mitigation plan, test/quality plan, and quality assurance
Md Tasnimul Hasan	Project Planner and Cost Estimator	Work Breakdown Structure (WBS), Gantt chart, resource planning, cost estimation, budget planning, progress tracking, and technical feasibility analysis

2 Software Requirements

2.1 Functional Requirements

1. FR1: The system shall allow patients to register and log in with a unique account.
2. FR2: The system shall allow patients to view available doctors and their schedules.
3. FR3: The system shall allow patients to book, reschedule, or cancel an appointment.
4. FR4: The system shall allow doctors to view and manage their daily appointment list.
5. FR5: The system shall send automatic confirmation and reminder notifications to patients.

2.2 Non-Functional Requirements

- NFR1: Only authorized users may access patient medical and appointment records.
- NFR2: The system should load appointment schedules within 2 seconds.
- NFR3: The interface should be clear and easy to navigate for first-time users.

2.3 Requirements Table

ID	Requirement	Priority
FR1	Patient Registration	High
FR2	View Doctor Schedules	High
FR3	Book/Cancel Appointment	High
FR4	Manage Doctor Schedule	High
FR5	Send Notifications	Medium

3 System Design

3.1 Use Case Diagram

Figure 1 shows the main use cases of the clinic appointment system. Patients can book appointments, while doctors can manage their schedules.

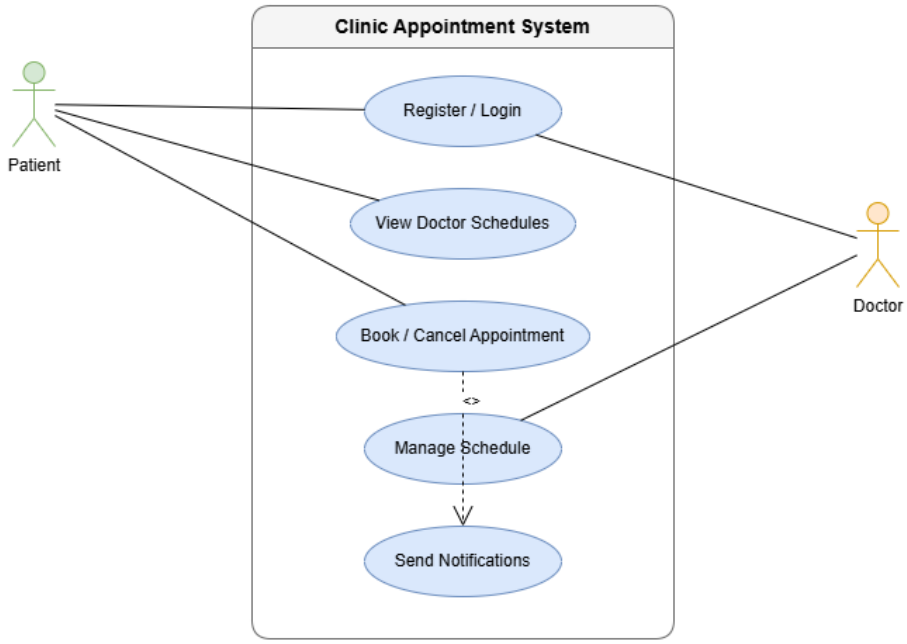


Figure 1: Use Case Diagram of Clinic Appointment System

3.2 Activity Diagram

Figure 2 shows the appointment booking workflow. An appointment is confirmed only if a time slot is available.

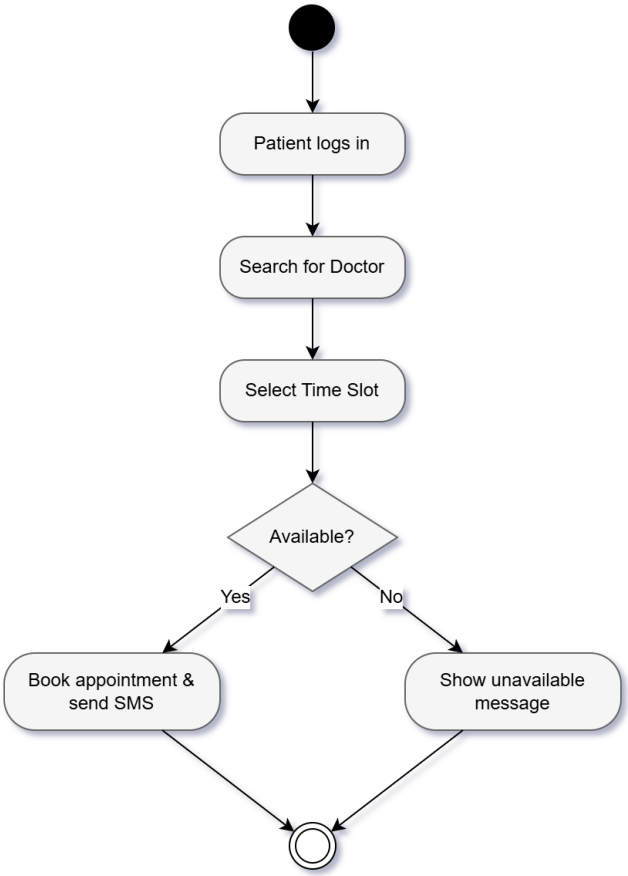


Figure 2: Appointment Booking Activity Diagram

3.3 Initial Class Diagram

Figure 3 presents five initial classes with attributes, methods, and relationships.

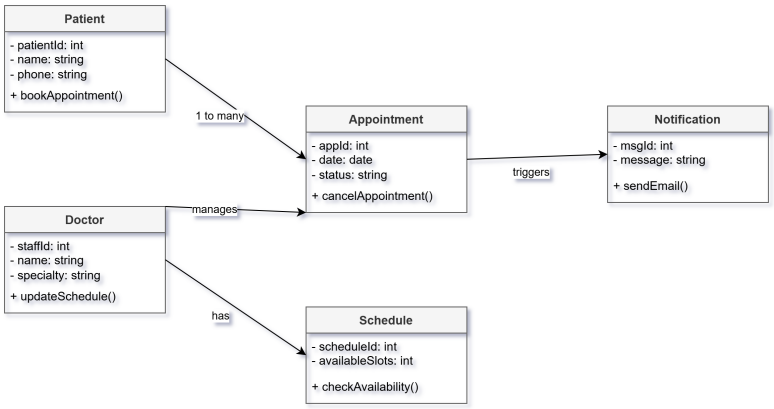


Figure 3: Initial Clinic Appointment System Class Diagram

4 Project Estimation

4.1 Effort and Time Estimation

Based on our initial requirements, the project is estimated to take a total of 120 working hours. The team consists of 4 members, and each member will contribute an average of 6 hours per week. Therefore, the estimated duration to complete the project is calculated as:

$$\text{Estimated Duration} = \frac{\text{Total Effort (120 hours)}}{\text{Team Capacity (4 members} \times \text{6 hours/week)}} = \frac{120}{24} = 5 \text{ weeks} \tag{1}$$

4.2 Project Completion

The overall progress of the project can be measured by calculating the percentage of completed tasks. Project completion is calculated as:

$$P = \frac{C}{T} \times 100 \tag{2}$$

where P is completion percentage, C is completed tasks, and T is total planned tasks. Equation 2 measures project progress. For example, if 45 out of 100 tasks are complete, then $P = \frac{45}{100} \times 100 = 45\%$.

5 Feasibility Study

5.1 Technical Feasibility

The system uses standard web technologies (HTML, CSS, JavaScript, and a MySQL database) which the team already knows. We have access to all necessary tools in the university computer lab.

5.2 Economic Feasibility

The project will use free open-source tools, so development cost is very low. Expected monthly hosting cost is around CNY 100.

The results suggest that the system will recover its initial cost within 2 years and provide a 50% return on investment.

5.3 Legal Feasibility

Patient data will be protected using encryption, following basic data privacy rules. No copyrighted software will be used.

5.4 Operational Feasibility

Clinic staff and patients can easily use the web interface without special training. The system will save time in managing appointments.

5.5 Schedule Feasibility

The team has 10 weeks to complete the project, which is enough to build the prototype and test it. **Decision:** GO. The project is technically possible, economically cheap, and legally safe.

5.6 ROI and Payback Period

Assumptions: Initial cost = CNY 20,000; Annual net benefit = CNY 10,000; Evaluation period = 3 years.

$$\text{ROI} = \frac{30000 - 20000}{20000} \times 100 = 50\% \tag{3}$$

$$\text{Payback} = \frac{20000}{10000} = 2 \text{ years} \tag{4}$$

6 Project Scope

6.1 In Scope

- Patient registration and secure login system.
- Doctor schedule viewing and time slot search.
- Booking, rescheduling, and cancelling appointments.
- Automatic SMS/Email notifications for appointments.

6.2 Out of Scope

- Online payment gateway for consultation fees.
- Video consultation feature.
- Online payment gateway for consultation fees.
- Video consultation feature.

Constraint: The system will only work on web browsers and not as a mobile app in this phase.

7 SMART Objectives

1. **Objective 1:** By Week 5, complete the patient booking module with 100% of planned test cases passing.
2. **Objective 2:** By Week 7, integrate the notification system (email/SMS) and ensure it sends alerts within 10 seconds of booking.
3. **Objective 3:** By Week 9, achieve a system response time of under 2 seconds for viewing doctor schedules during a load test of 50 users.

8 Work Breakdown Structure

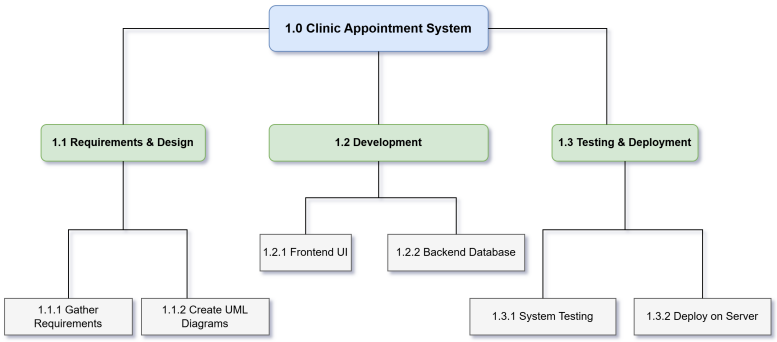


Figure 4: Work Breakdown Structure for Clinic Appointment System

9 WBS Dictionary

WBS Code	Deliverable Name	Effort (hrs)	Acceptance Criterion
1.2.1	Patient Login Module	15	User can log in with valid credentials.
1.2.2	Doctor Schedule View	20	Displays available slots accurately.
1.3.1	Booking System Test	10	System correctly books and sends email.